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B.Tech. Degree V Semester Examination November 2015

ME 1506 POWER PLANT ENGINEERING (2012 scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) What is the significance of incremental rate for a power plant?
 (b) Explain briefly load curve and load duration curve.
 (c) State the advantages of gas turbine power plant over diesel power plant.
 (d) What factors should be considered while selecting a site for diesel power plant?
 (e) What are the advantages of FBC system over the conventional one?
 (f) What are the principal reasons for operating steam turbines with condenser?
 (g) Give classifications of a tidal power plant and list out its components.
 (h) Write a short note on "Wind Electricity Economics".

PART B

(4 × 15 = 60)

- II. Determine the following from the daily load curve as per the table below if (3 × 5 = 15) the max. demand of a power station is 96000KW.

Time hours	0-6	6-8	8-12	12-14	14-18	18-22	22-24
Load (MW)	48	60	72	60	84	96	48

- (i) Determine the load factor of power station.
 (ii) What is the load factor of standby equipment rated at 30 MW that takes up all load in excess of 72 MW?
 (iii) Calculate its use factor.

OR

- III. (a) Explain briefly various methods used to calculate the depreciation cost. (9)
 (b) How can power generation cost be reduced? (6)
- IV. Explain the essential components of a diesel power plant with sketches. (15)

OR**(P.T.O.)**

- V. (a) What are the major fields of application and limitations of gas turbine power plants? (9)
- (b) Briefly explain closed cycle gas turbine power plant. (6)
- VI. (a) Discuss suitability of steam plant to supply load in an interconnected system. (8)
- (b) Why are multiple chimneys preferred to single unit chimneys? (7)
- OR**
- VII. (a) List out the advantages and disadvantages of a nuclear plant over conventional thermal plants. (7)
- (b) What are the main purpose of super heater? What are the advantages of superheated steam? (8)
- VIII. (a) Describe working of a solar power plant by using a neat sketch. (8)
- (b) Explain working of a fuel cell and list its advantages over other non conventional systems of power generation. (7)
- OR**
- IX. (a) Describe the working of a geothermal power plant. (8)
- (b) Discuss the merits and limitations of a wind mill power generation unit. (7)
